

Metal Suppliers Online: Material Property Data

Stainless Steel Alloy 455

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Availability

Stainless Steel Alloy 455 is stocked by 19 North American distributors and produced by 13 large mills. Distributors will offer small quantity buys while mills will generally only sell large quantities, with delivery times anywhere from 10 to 50 weeks depending on size and form required

This material is stocked primarily in Bar Products by 17 distributors but is also available to a lesser extent in Tubular Products, Forging Products, Wire Products, and Flat Rolled Products.

For distributor (small quantity) availability click [here](#)

For mill (large volume) production click [here](#)

To view all click [here](#)

Specifications

The following specifications cover **Stainless Steel Alloy 455**

- | | |
|-------------------------------|-------------------------------|
| • AMS 5578 | • AMS 5617 |
| • AMS 5672 | • AMS 5860 |
| • ASTM A313 | • ASTM A564 |
| • ASTM A693 | • ASTM A705 |
| • MIL F-83311 | • MIL S-83311 |
| • UNS S45500 | |

Property Results

Related Metals:

- Custom 455(tm)
- HPM 455(tm)

Chemistry Data : [\[top\]](#)

Carbon	0.05 max
Chromium	11 - 12.5
Copper	1.5 - 2.5
Iron	Balance
Manganese	0.5 max
Molybdenum	0.5 max
Nickel	7.5 - 9.5
Niobium	0.1 - 0.5
Phosphorus	0.04 max
Silicon	0.5 max
Sulphur	0.03 max
Titanium	0.8 - 1.4

Principal Design Features

This alloy is a martensitic stainless that combines corrosion resistance similar to 304, easy fabrication and one step age hardenability yielding strength characteristics similar to martensitic stainless'. It possesses a yield strength that is roughly 3 times the strength of standard 304 stainless. This alloy is extremely stable during heat treating so it can be machined to close tolerance prior to aging.

Applications

Generally employed in applications where standard austenitics fail in terms of strength requirements in corrosive environments and martensitics fail in terms of corrosion resistance. It effectively brings together the best of both worlds.

Machinability

Utilize speeds and feeds comparable with other martensitic stainless steels. Positive feeds and slow speeds will yield best results.

Welding

It is not necessary to preheat this material and both resistance and shielded fusion methods have been commonly employed. It is recommended, for best properties, that material be aged after welding, although this may not be necessary depending on the end use.

Hot Working

Soak material at 2100-2150 F(1150-1177 C), air cool after working and fully anneal.

Cold Working

With a relatively low work hardening rate, this alloy may be effectively cold worked using most common methods.

Annealing

Soak thoroughly at 1875-7925 F(1024-1052 C) for 1 hour, water or oil quench.

Aging

To obtain condition H900, H950, H1000, H1050 or H1150, soak material at desired temperature for 4 hours and air cool.

Physical Data : [\[top\]](#)

Density (lb / cu. in.)	0.28
Specific Gravity	7.75
Specific Heat (Btu/lb/Deg F - [32-212 Deg F])	0.11
Electrical Resistivity (microhm-cm (at 68 Deg F))	597
Modulus of Elasticity Tension	28

Mechanical Data : [\[top\]](#)

MSO currently has no data available for this grade.

Videos :

MSO currently has no videos available for this grade.

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